HURST NEURAL OPERATOR

TRADING STRATEGY

Neural Operators · Physics-Informed ML · SPDEs · HPC

Strategy Development Plan | Target Sharpe ≥ 1.25

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TARGET SHARPE	ASSET CLASS	FREQUENCY	COMPUTE
			Local GPU

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1 | EXECUTIVE SUMMARY

This document presents a comprehensive development plan for a systematic futures trading strategy centred on a **time-varying Hurst exponent** predicted in real time by a **Fourier Neural Operator (FNO)**. The Hurst exponent $H \in (0, 1)$ governs the memory structure of a price process: values above 0.5 imply trending (persistent) dynamics, values below 0.5 imply mean-reverting (anti-persistent) dynamics, and values near 0.5 approximate a random walk. By accurately estimating $H(t)$ and switching regime-conditional sub-strategies in real time, the system aims to harvest alpha across a wide range of market environments.

Four technical pillars differentiate this approach from classical signal strategies: (i) a neural operator that treats Hurst estimation as an *operator learning* problem mapping function spaces rather than fixed-dimensional vectors; (ii) physics-informed loss terms derived from fractional Brownian motion (fBm) theory; (iii) a stochastic PDE (SPDE) simulator used for synthetic data augmentation and risk-scenario generation; and (iv) a high-performance parallel computing stack (multi-GPU PyTorch + CUDA kernels) that makes the pipeline feasible on a local GPU workstation.

Preliminary back-of-envelope estimates place the strategy's Sharpe ratio target at ≥ **1.25 annualised** on a diversified futures universe (equity index, commodity, and rates futures), contingent on careful transaction-cost modelling and live execution latency management.

2 | INVESTMENT THESIS & ASSET CLASS SELECTION

Empirical market microstructure research has consistently demonstrated that price series do *not* possess a constant Hurst exponent. The exponent evolves with liquidity regimes, macro uncertainty, and participant composition. Strategies that ignore this non-stationarity — applying fixed trend-following or mean-reversion rules — suffer precisely during transitions. This strategy explicitly models and profits from those transitions.

Why Futures?

- Leverage and capital efficiency: margin requirements allow broad diversification at modest notional.
- Continuous roll structure provides long price histories ideal for SPDE calibration.
- CME/ICE liquidity depth minimises market impact at typical position sizes.
- Fractional dynamics are well-documented in equity index, energy, and rates futures.
- No short-sale constraints — the strategy must go both long and short.

Recommended Universe

Sector	Instruments	Rationale
Equity Index	ES, NQ, RTY (CME)	Strong fractal structure; high H(t) variance
Commodities	CL, GC, NG (NYMEX)	Supply shocks create pronounced H shifts
Rates	ZN, ZB (CBOT)	Mean-reversion regimes well-documented
FX Futures	6E, 6J, 6B (CME)	Regime diversity; low correlation to equity

Recommended Frequency: Daily-bar signal generation with intraday execution. Hurst estimation windows of 60–252 daily bars yield statistically stable estimates; intraday execution (via TWAP/VWAP algorithms) keeps slippage manageable.

3 | MATHEMATICAL FOUNDATIONS

3.1 Fractional Brownian Motion & the Hurst Exponent

A fractional Brownian motion $B^H(t)$ with Hurst parameter $H \in (0,1)$ is the unique (up to scaling) Gaussian process satisfying:

$$E[\ B^H(t)\ B^H(s)\ ] = (1/2)(|t|^{2H} + |s|^{2H} - |t-s|^{2H})$$

When $H = 0.5$ the increments are independent (standard BM). For $H > 0.5$ increments are positively correlated (long-range dependence / trending). For $H < 0.5$ they are negatively correlated (anti-persistent / mean-reverting). The spectral density of fBm increments follows a power law:

$$S(f) \propto |f|^{-(2H+1)}$$

This power-law structure is what the neural operator exploits: it learns the mapping from the empirical spectrum of a price window to a function-valued estimate of $H(t)$.

3.2 Stochastic PDE Framework

We model log-price dynamics under the rough fractional SPDE:

$$d \log S(t) = \mu(t)\ dt + \sigma(t)\ dB^{H(t)}(t)$$

where $\sigma(t)$ itself follows a rough Bergomi-style volatility process:

$$d \log \sigma(t) = \kappa(\theta - \log \sigma(t))\ dt + \nu \int\blacksquare\blacksquare\ (t-s)^{H-1/2}\ dW(s)$$

Key properties and uses of this SPDE framework within the strategy:

- **Calibration:** Fit (μ, $\sigma\blacksquare$, κ, θ, ν, H) to rolling windows of market data using maximum likelihood on the fractional covariance structure.
- **Simulation:** Use fast GPU-parallelised Cholesky or Hosking's method to generate 10,000+ synthetic price paths per instrument per day for risk scenario analysis.
- **Physics loss:** SPDE residuals are embedded as soft constraints in the FNO training loss, preventing the neural operator from predicting H values inconsistent with observed spectral density.
- **Option hedging:** Under rough volatility ($H < 0.5$), standard Black-Scholes hedges are biased — SPDE calibration allows correct delta/vega hedging of any attached options overlay.

3.3 Regime Classification via H(t)

The predicted $H(t)$ maps directly to a discrete regime label used by the signal engine:

Regime	H(t) Range	Interpretation	Strategy Action
TREND	$H > 0.60$	Persistent, long memory	Momentum long/short
NEUTRAL	$0.45 \le H \le 0.60$	Near-random walk	Reduce gross exposure
REVERSION	$H < 0.45$	Anti-persistent, mean-reverting	Stat-arb / pairs
TRANSITION	$\Delta\|H\| > 0.08$ / day	Rapid regime shift	Flatten + reassess

4 | NEURAL OPERATOR ARCHITECTURE

4.1 Fourier Neural Operator (FNO) for Hurst Prediction

Classical Hurst estimators (R/S analysis, DFA, Whittle MLE) are computed on fixed windows and produce a single scalar. The FNO instead treats the mapping from *price-return function* to *Hurst-exponent function* as an operator between infinite-dimensional Banach spaces, discretisation-invariant and resolution-adaptive.

The FNO operator G: a(x) → H(x) is defined via a series of Fourier integral layers:

```
v_j+1(x) = σ( W v_j(x) + F⁻¹[ R_j · F[v_j] ](x) )
```

where F / F^{-1} denote the discrete Fourier transform pair, R_j are learnable complex-valued weight tensors truncated to the k_max lowest Fourier modes, W is a local linear transform, and σ is a pointwise nonlinearity (GELU).

Input representation:

- Rolling windows of log-returns at multiple resolutions: [5, 21, 63, 126, 252] bars.
- Realised variance and bipower variation estimates.
- Order-flow imbalance (OFI) and signed volume features.
- Cross-instrument correlation matrix (flattened upper triangle).
- Calendar features: day-of-week, month, VIX percentile rank.

Output representation:

A function H: [0, T] → (0, 1) giving the predicted Hurst exponent at each point in the input time grid. At inference time, the value H(T) (current estimate) is used for regime classification; the full trajectory is used for uncertainty quantification (width of the H(t) confidence band modulates position sizing).

Architecture hyperparameters (baseline):

Parameter	Value
FNO layers (depth)	4
Fourier modes k_max	32 per dimension
Hidden channel width	64
Activation	GELU
Input sequence length	252 daily bars
Output length	252 (same grid)
Parameters (total)	~2.1 M
Training epochs	300 with cosine-annealing LR
Batch size	256 (multi-GPU data-parallel)

4.2 Physics-Informed Loss Functions

The total training loss combines a data-fitting term, a physics residual, and a smoothness regulariser:

$$L\_total = \lambda\_data \cdot L\_data + \lambda\_phys \cdot L\_phys + \lambda\_smooth \cdot L\_smooth$$

L_data — Supervised Hurst MSE:

Ground-truth H values are generated by Whittle MLE on synthetic fBm paths (known H by construction). The supervised loss is the mean-squared error between the FNO prediction and the Whittle estimate over the training set.

L_phys — Spectral Power-Law Residual:

For each predicted H(t), the implied log-power-spectrum slope should satisfy $\beta = -(2H+1)$. The physics loss penalises deviation between the FNO-implied spectral slope and the empirically measured periodogram slope of the corresponding window:

$$L\_phys = || \beta\_predicted - \beta\_empirical ||^2 \text{ where } \beta\_predicted = -(2\ H\_FNO + 1)$$

L_smooth — Temporal Sobolev Regulariser:

Penalises high-frequency oscillations in H(t) using a first-order Sobolev semi-norm, preventing the operator from predicting non-physical rapid regime flickers:

$$L\_smooth = || dH/dt ||^2_{L2}$$

> Recommended initial weights: λ_data = 1.0, λ_phys = 0.3, λ_smooth = 0.05. These are tuned via a grid search on a held-out validation set.

4.3 Training Pipeline

- **Synthetic pre-training (Phase A):** Generate 500 k fBm paths with H ~ Uniform(0.1, 0.9) via GPU-parallelised Hosking method. Train FNO to convergence (~2 h on a 4×RTX 3090).
- **Real-data fine-tuning (Phase B):** Fine-tune on historical futures data (2000–2024) using pseudo-labels from Whittle MLE. Learning rate 10× lower than pre-training.
- **Online adaptation (Phase C):** Daily gradient step on the most recent 63-bar window to track structural breaks. Exponential moving average of model weights for stability.
- **Uncertainty quantification:** MC-Dropout at inference (T=50 forward passes) gives a distribution over H(t), enabling confidence-weighted position sizing.

5 | SIGNAL GENERATION & PORTFOLIO CONSTRUCTION

The signal pipeline converts the FNO's H(t) output into signed position weights for each instrument in the universe. Two parallel sub-strategies run simultaneously; the portfolio allocator blends them according to the current regime.

Sub-strategy A — Trend Following (H > 0.60)

- Signal: sign(EMA_fast – EMA_slow) × |H – 0.5| scaling, where EMA windows are chosen adaptively as a function of H (higher H → wider windows).
- Entries triggered when H crosses 0.60 upward AND price breaks a Donchian channel.
- Stop: trailing ATR(14) stop, widened proportionally to σ(H(t)) (FNO uncertainty).

Sub-strategy B — Mean Reversion (H < 0.45)

- Signal: z-score of price vs. 20-bar rolling mean, threshold ±1.5σ.
- Enter long (short) when z < –1.5 (z > +1.5) AND H < 0.45.

- Exit when z reverts to 0 or H rises above 0.50 (regime switch).
- Cross-instrument: pairs formed by clustering instruments by H-trajectory similarity (DTW distance), enabling stat-arb within regimes.

Portfolio Allocator

Position weights w_i(t) are computed via a hierarchical risk parity (HRP) optimisation over the SPDE-simulated covariance matrix, with an overlay that scales gross exposure by the regime confidence score:

```
w_i(t) = w_i_HRP(t) × confidence(H_i(t)) × vol_target / σ_portfolio(t)
```

Target annualised portfolio volatility: 10–15%. Maximum instrument weight: 20% of NAV. Maximum sector concentration: 40% of NAV.

Transaction Cost Model

Costs are modelled as: TC = spread/2 + market_impact_coefficient × √(Q / ADV) where Q is order size and ADV is average daily volume. Signals with expected gross alpha below 2× estimated TC are suppressed (cost filter).

6 | RISK MANAGEMENT

Risk Dimension	Control Mechanism
Position-level stop	Trailing ATR(14) × (1 + σ_H) — widens in uncertain regimes
Portfolio drawdown	Gross exposure halved when 10-day drawdown exceeds 3%; reset at new equity high
Regime-transition risk	All positions flattened within 2 bars when |ΔH| > 0.08/day
Model risk (FNO)	Shadow classical estimator (DFA) runs in parallel; alert if |H_FNO – H_DFA| > 0.15
Concentration risk	Max 20% single instrument, 40% per sector, HRP rebalancing daily
Leverage cap	Max 4× notional; de-lever to 2× in NEUTRAL regime
Tail risk	5% of gross margin reserved for OTM put spreads on index futures
Execution risk	TWAP slicing over 30-min window; abort if fill rate < 60% in 10 min
Data quality	Z-score anomaly detector on raw prices; auto-quarantine on 5σ event

7 | HIGH-PERFORMANCE PARALLEL COMPUTING STACK

The strategy's pipeline is computationally intensive at three points: FNO training, SPDE Monte Carlo simulation, and daily inference across the full universe. The following HPC architecture targets a local GPU workstation (recommended: 2–4× NVIDIA RTX 4090 / A6000 GPUs).

Layer 1 — GPU-Parallelised SPDE Simulation

- Hosking's exact method for fBm simulation is inherently sequential. We instead use the **circulant embedding + FFT** approach (Davies-Harte algorithm): O(N log N) per path, trivially parallelised across paths on GPU.
- PyTorch custom CUDA kernel batches 65,536 fBm paths simultaneously on a single A6000.

- Used for: synthetic training data generation (offline), daily risk scenario runs (online).

Layer 2 — Distributed FNO Training (PyTorch DDP)

- PyTorch DistributedDataParallel across all available GPUs using NCCL backend.
- Mixed precision (bfloat16) for forward/backward passes; float32 master weights.
- Gradient checkpointing to fit large batch sizes within VRAM budget.
- Expected training time: ~2 h for 300 epochs on 4× RTX 4090 (synthetic pre-training).

Layer 3 — Real-Time Inference Pipeline

- TorchScript / ONNX export of trained FNO for deployment latency < 50 ms per universe scan.
- Async data ingestion via ZeroMQ; price updates trigger incremental feature recomputation.
- MC-Dropout uncertainty pass (50 samples) executes in parallel on single GPU stream.
- Total end-of-day computation: ~3 min on RTX 4090 for 12-instrument universe.

Layer 4 — Backtesting Parallelism

- Walk-forward backtests use Python multiprocessing (one worker per parameter set / fold).
- Joblib with loky backend; pinning workers to physical cores avoids NUMA effects.
- Vectorised portfolio accounting in NumPy/Polars; avoids Python loops over bar history.

> Recommended local hardware: 2–4× NVIDIA RTX 4090 (24 GB VRAM each), AMD Threadripper or Intel Xeon w-series CPU (24+ cores), 128 GB DDR5 RAM, NVMe SSD RAID for tick storage. Total build cost: ~USD 12,000–18,000.

8 | BACKTESTING & VALIDATION METHODOLOGY

Robust out-of-sample validation is essential because the FNO is trained on historical data and could be subject to look-ahead bias if not handled carefully.

Expanding-window Walk-Forward

The backtest uses a strict expanding-window protocol: the model is trained on all data up to date T, then evaluated on the subsequent 252 bars, rolling forward annually. No future data is ever used in feature construction or model selection.

Purging & Embargo

Due to the rolling-window features (up to 252-bar lookback), the training set is purged of the 252 bars immediately preceding each test period, and an additional 21-bar embargo is applied. This prevents leakage via feature overlap.

Combinatorial Purged Cross-Validation (CPCV)

López de Prado's CPCV with k=10 splits and backtest paths p=2 provides an unbiased estimate of the Sharpe distribution and confidence interval. Target: lower 5th-percentile Sharpe > 0.80 (conservative floor).

Key Validation Statistics

Metric	Target	Notes
Annualised Sharpe Ratio	≥ 1.25	Net of costs, CPCV median estimate
Annualised Return	≥ 15%	On 10% vol target

Max Drawdown	≤ 18%	Calendar-day basis
Calmar Ratio	≥ 0.80	Return / Max DD
Win Rate	45–55%	Higher not required given risk/reward
Profit Factor	≥ 1.4	Gross profit / gross loss
H-prediction MAE	≤ 0.04	Mean absolute error on test set
Regime classification F1	≥ 0.72	3-class: TREND / NEUTRAL / REVERSION
Turnover (annualised)	≤ 8× NAV	Cost-effective execution

Stress Tests

- 2008 GFC, 2020 COVID crash, 2022 rate shock: measure drawdown and recovery time.
- Transaction cost sensitivity: stress TC by 3× baseline to test robustness.
- H-estimator degradation: replace FNO with DFA classical estimator; compare Sharpe.
- Regime-label noise injection: flip 10% of regime labels; measure signal degradation.

9 | IMPLEMENTATION ROADMAP

The strategy development is structured as six sequential phases over approximately nine months, each with a clear gate criterion before proceeding.

Phase	Name	Duration	Key Deliverable
1	Data Infrastructure	6 weeks	Historical futures data (Refinitiv/Bloomberg), tick + daily bars, clean & store in Parquet/Arrow; SPDE simulator passing statistical tests for fBm properties
2	Classical Hurst Baseline	4 weeks	R/S, DFA, and Whittle MLE estimators implemented and validated on synthetic fBm; regime classification baseline Sharpe ≥ 0.70 established
3	FNO Development & Pre-Training	8 weeks	FNO architecture implemented in PyTorch; synthetic pre-training complete; H-prediction MAE < 0.04 on held-out synthetic test set
4	PINN Fine-Tuning on Real Data	6 weeks	Physics-informed loss integrated; fine-tuning on 2000–2024 futures data; regime classification F1 ≥ 0.72 on walk-forward validation
5	Signal & Portfolio Engine	6 weeks	Full signal pipeline, HRP allocator, TC model, and risk overlays implemented; walk-forward backtest Sharpe ≥ 1.25; max drawdown ≤ 18%

6	Paper Trading & Live Deployment	8 weeks	6-week paper trade on IB TWS / Interactive Brokers API; live performance within 15% of backtest Sharpe; TWAP execution latency < 500 ms

Gate criteria: Each phase gate requires sign-off on the key deliverable before proceeding. Phases 3–5 may run in partial parallel (FNO training while classical baseline strategy is being stress-tested).

10 | PERFORMANCE TARGETS & SUCCESS CRITERIA

Criterion	Go-Live Threshold	Stretch Target
CPCV Median Sharpe	≥ 1.25	≥ 1.60
Worst CPCV Path Sharpe	≥ 0.80	≥ 1.00
Max Drawdown	≤ 18%	≤ 12%
Calmar Ratio	≥ 0.80	≥ 1.20
Paper Trade 6-wk Sharpe	≥ 0.90 annualised	≥ 1.25 annualised
H-prediction MAE	≤ 0.04	≤ 0.025
Regime F1 score	≥ 0.72	≥ 0.82
End-of-day latency	≤ 5 min	≤ 60 sec

11 | KEY RISKS & MITIGATIONS

Risk	Likelihood	Impact	Mitigation
FNO overfits to in-sample H dynamics	Medium	High	Strict CPCV; synthetic pre-training; ensembling with classical estimator
Regime labels are noisy / ambiguous	High	Medium	Soft regime scores (probabilities) rather than hard labels; confidence threshold
Transaction costs erode alpha	Medium	High	Cost filter (suppress signals < 2× TC); parameter robustness testing
SPDE model misspecification	Medium	Medium	Multiple SPDE variants tested; mGAN augmentation as robustness check
GPU hardware failure	Low	Medium	Daily model checkpoints; fallback to DFA classical estimator
Market microstructure change	Low	High	Online fine-tuning (Phase C); structural break detector on H time series
Data vendor outage	Low	Medium	Dual vendor redundancy (primary + Quandl/yfinance backup)